AZ-104

Administer Governance and Compliance

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AZ-104 Agenda

LP1

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

Learning Objectives

- Describe the core architectural components of Azure
- Azure Policy Initiatives
- Secure your Azure resources with role-based access control (RBAC)
- Lab 02a - Manage Subscriptions and RBAC
- Lab 02b - Manage Governance via Azure Policy

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Describe the core architectural components of Azure

Learning Objective – Subscriptions and Azure RM

- Identify Regions
- Implement Azure Subscriptions
- Identify Subscription Usage
- Obtain a Subscription
- Create Resource Groups
- Determine Service Limits and Quotas
- Create an Azure Resource Hierarchy
- Apply Resource Tagging
- Manage Costs
- Learning Recap

Exam objectives

Manage subscriptions and governance

- Configure resource locks
- Apply and manage tags on resources
- Manage resource groups
- Manage subscriptions
- Manage costs by using alerts, budgets, and Azure Advisor recommendations
- Configure management groups

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Identify Regions

1 Region = 3 Data Center
Avail Zone

67 Regions
westeurope
northeurope

A region represents a collection of datacenters

Provides flexibility and scale

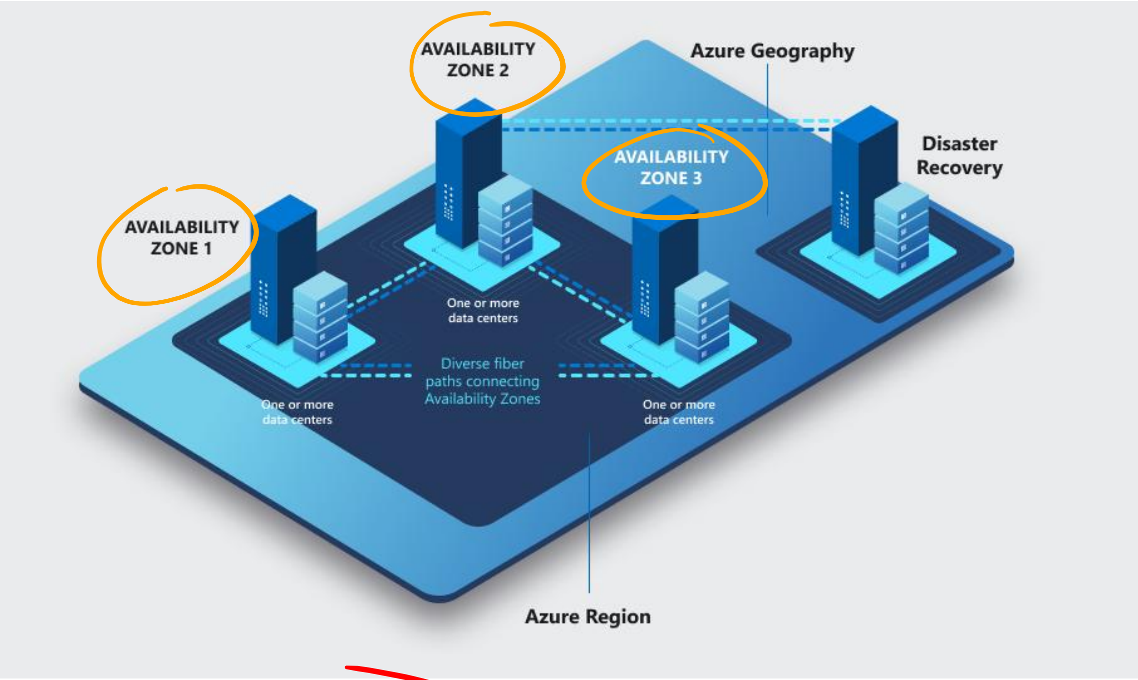
Preserves data residency

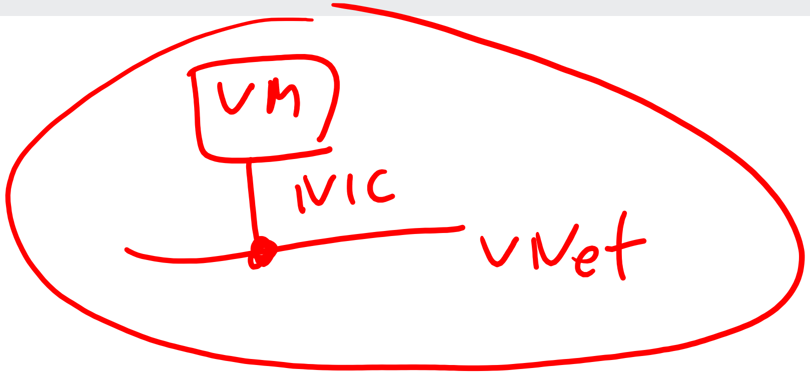
Select regions close to your users

Be aware of region deployment availability

There are global services that are region independent

Most regions are paired for high availability





Region

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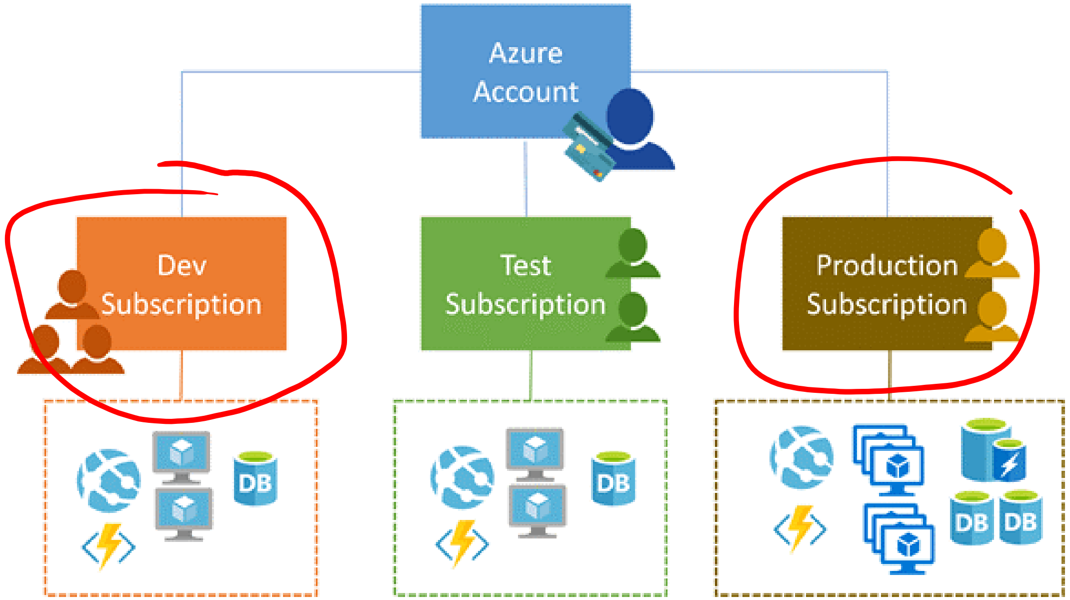
Implement Azure Subscriptions

- Only identities in Entra ID, or in a directory that is trusted by Entra ID, can create a subscription
- Logical unit of Azure services that is linked to an Azure account
- Security and billing boundary



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Identify Subscription Usage

Subscription	Usage
Free	€ Includes a $200 credit for the first 30 days, free limited access for 12 months CC
Pay-As-You-Go	Charges you monthly
CSP	Agreement with possible discounts through a Microsoft Cloud Solutions Provider Partner – typically for small to medium businesses
Enterprise	One agreement, with discounts for new licenses and Software Assurance – targeted at enterprise-scale organizations
Student	Includes $100 for 12 months – must verify student access

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Obtain a Subscription

Enterprise Agreement customers make an upfront monetary commitment and consume services throughout the year

Resellers provide a simple, flexible way to purchase cloud services

Partners can design and implement your Azure cloud solution

Personal free account – Start right away

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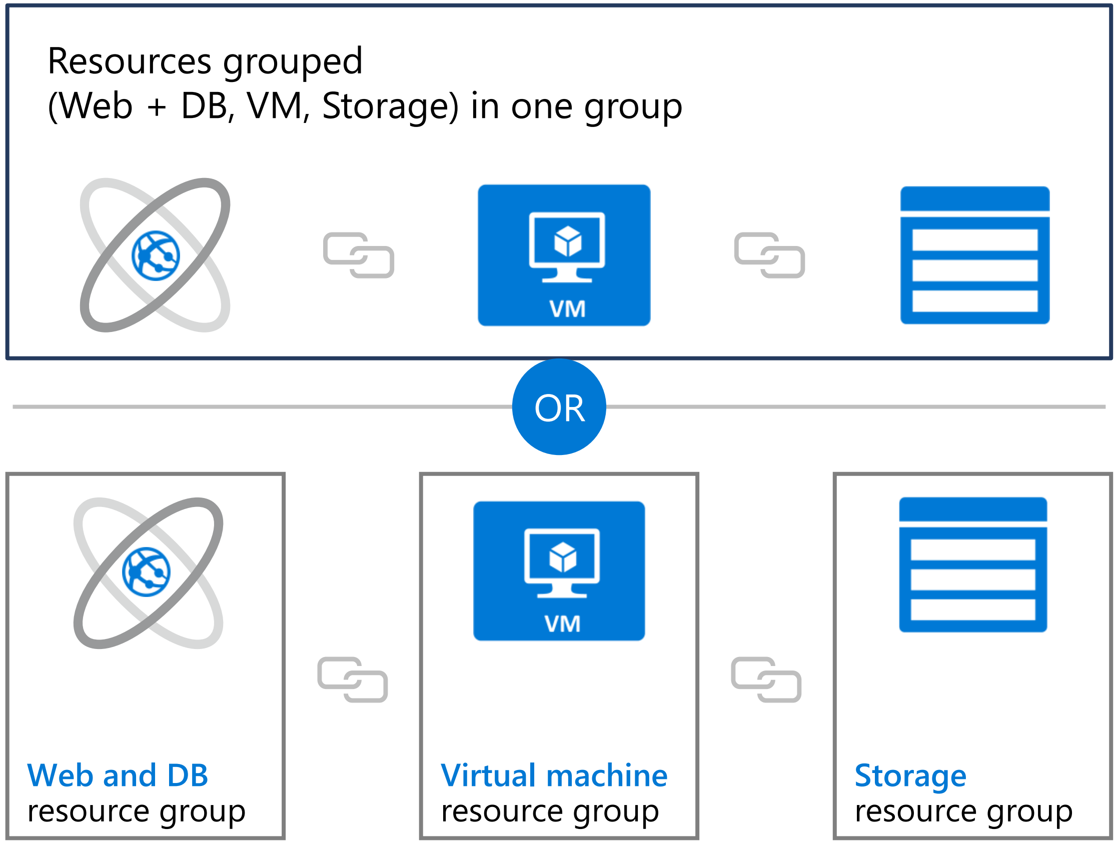
Create Resource Groups

- Resources can only exist in one resource group
- Groups can have resources of many different types (services) and from many different regions
- Groups cannot be renamed or nested
- You can move resources between groups



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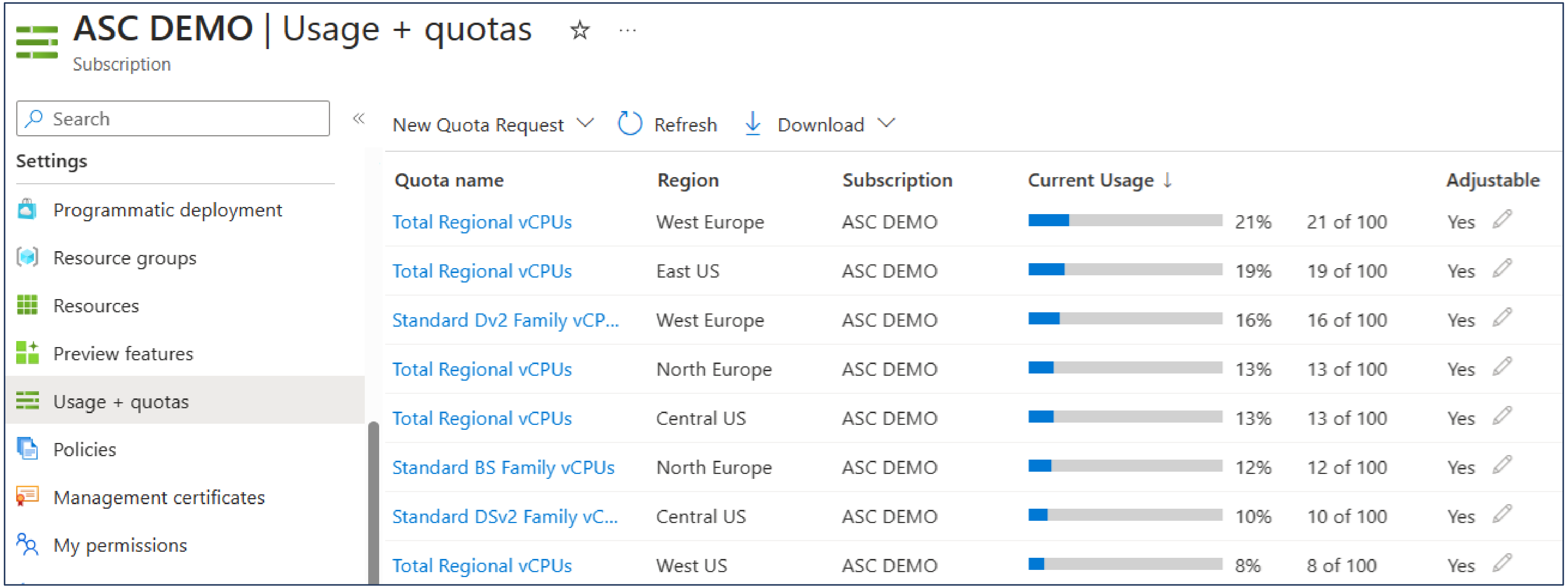
Determine Service Limits and Quotas



Resources have a default limit - a subscription quota

Helpful to track current usage, and plan for future use

You can open a free support case to increase limits to published maximums

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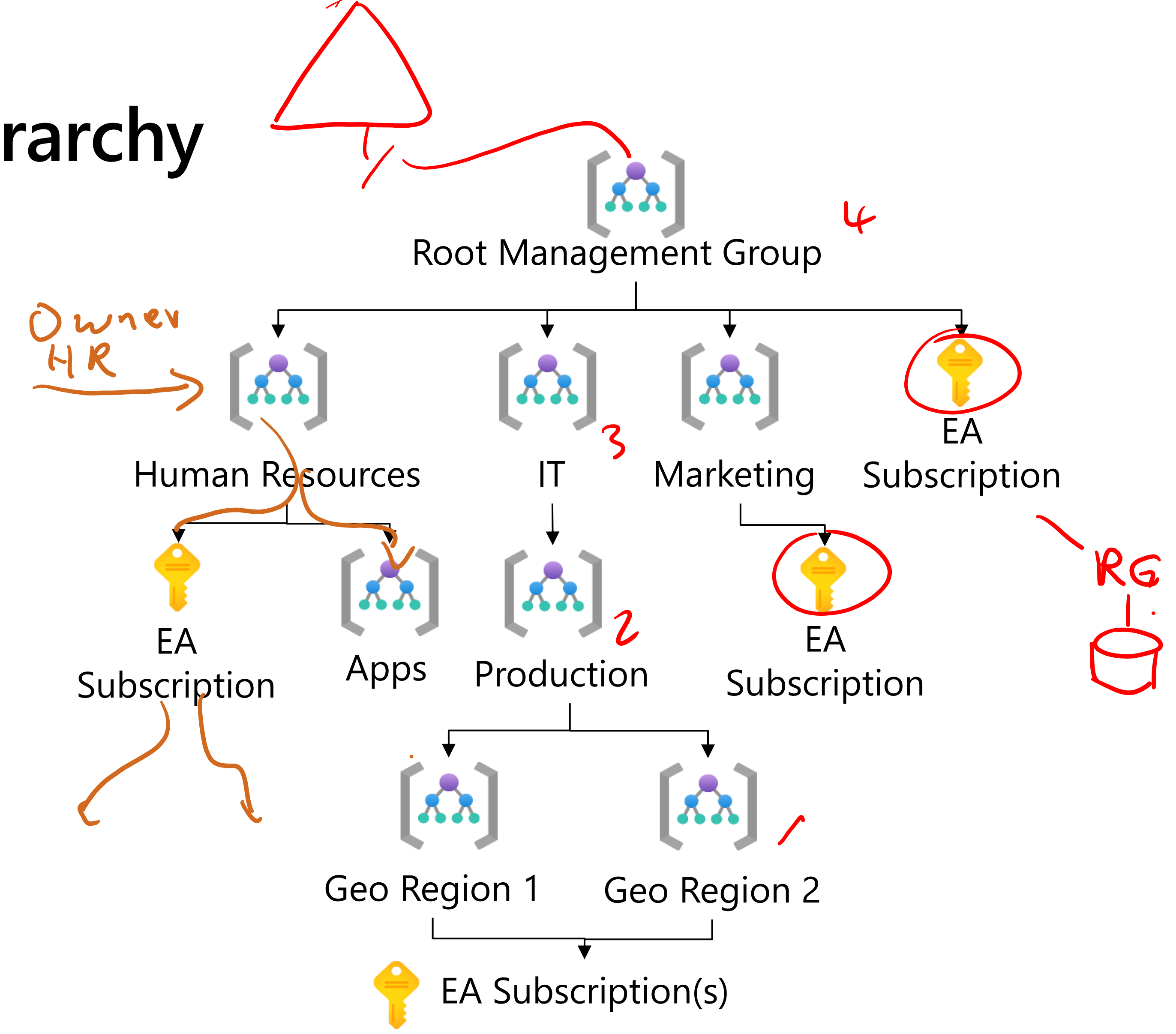
Create an Azure Resource Hierarchy

- Management groups provides a level of scope above subscriptions
- Target policies and spend budgets across subscriptions and inheritance down the hierarchies
- Implement compliance and cost reporting by organization (business/teams)



* To prevent changes, apply resource locks at the subscription, resource group, or resources level

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Apply Resource Tagging

RG

Tag

Costc : 4711

cost 4711 VM

Policy

- Provides metadata for your Azure resources
- Logically organizes resources
- Consists of name-value pair
- Useful for rolling up billing information

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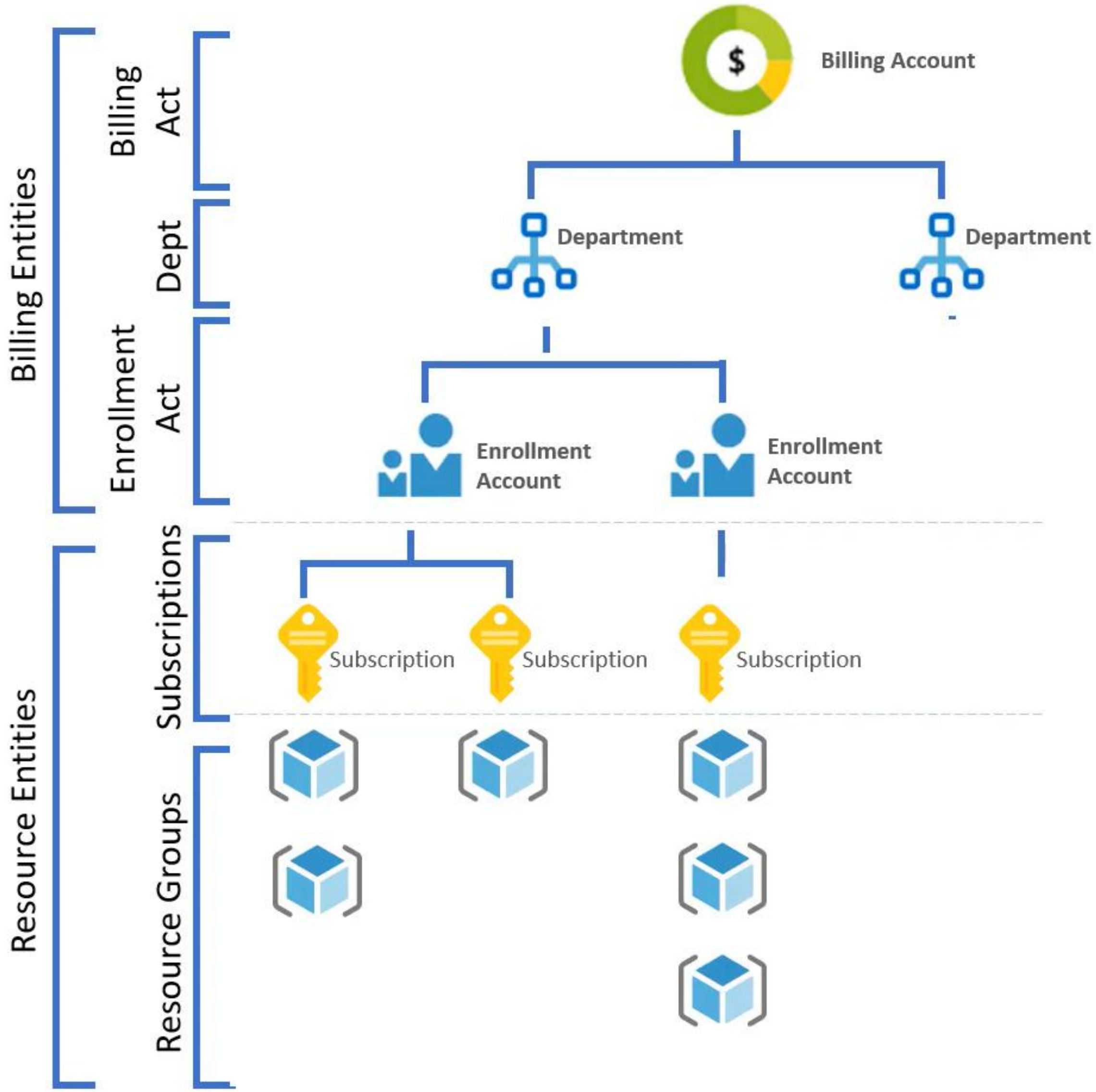
Manage Costs

- Costs are resource-specific
- Usage costs may vary between locations
- Costs for inbound and outbound data transfers differ
- Pre-pay with Azure reserved instances
- Use your on-premises licenses with Azure Hybrid Benefit
- Optimize with alerts, budgets, and Azure Advisor recommendations



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Azure Policy Initiatives

Learning Objective – Azure Policy initiatives

- Implement Azure Policy
- Implement Azure Policies
- Create Azure Policies
- Demonstration – Azure Policy
- Learning Recap

Exam objectives

Manage subscriptions and governance

- Implement and manage Azure Policy

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Implement Azure Policies

- A service to create, assign, and manage policies
- Runs evaluations and scans for non-compliant resources

Advantages:

- Enforcement and compliance
- Apply policies at scale
- Remediation

Usage Cases
Allowed resource types – Specify the resource types that your organization can deploy
Allowed virtual machine SKUs – Specify a set of virtual machine SKUs that your organization can deploy
Allowed locations – Restrict the locations your organization can specify when deploying resources
Require tag and its value – Enforces a required tag and its value
Azure Backup should be enabled for Virtual Machines – Audit if Azure Backup service is enabled for all Virtual machines

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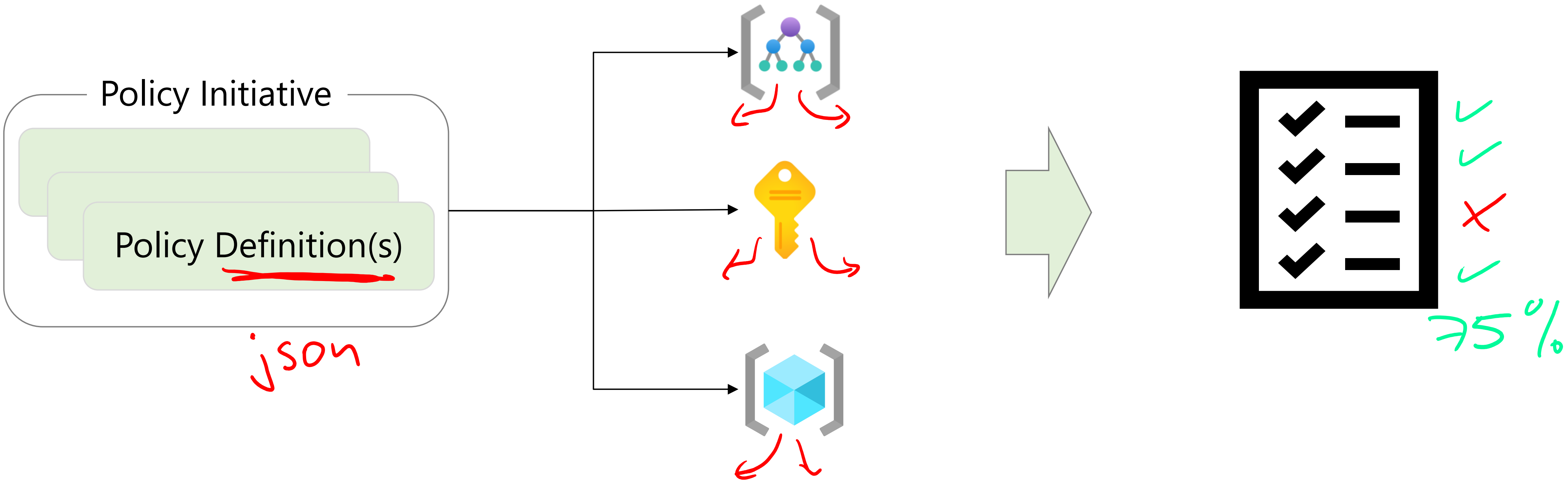
Create Azure Policies

Define and create

Scope and assign

Assess compliance



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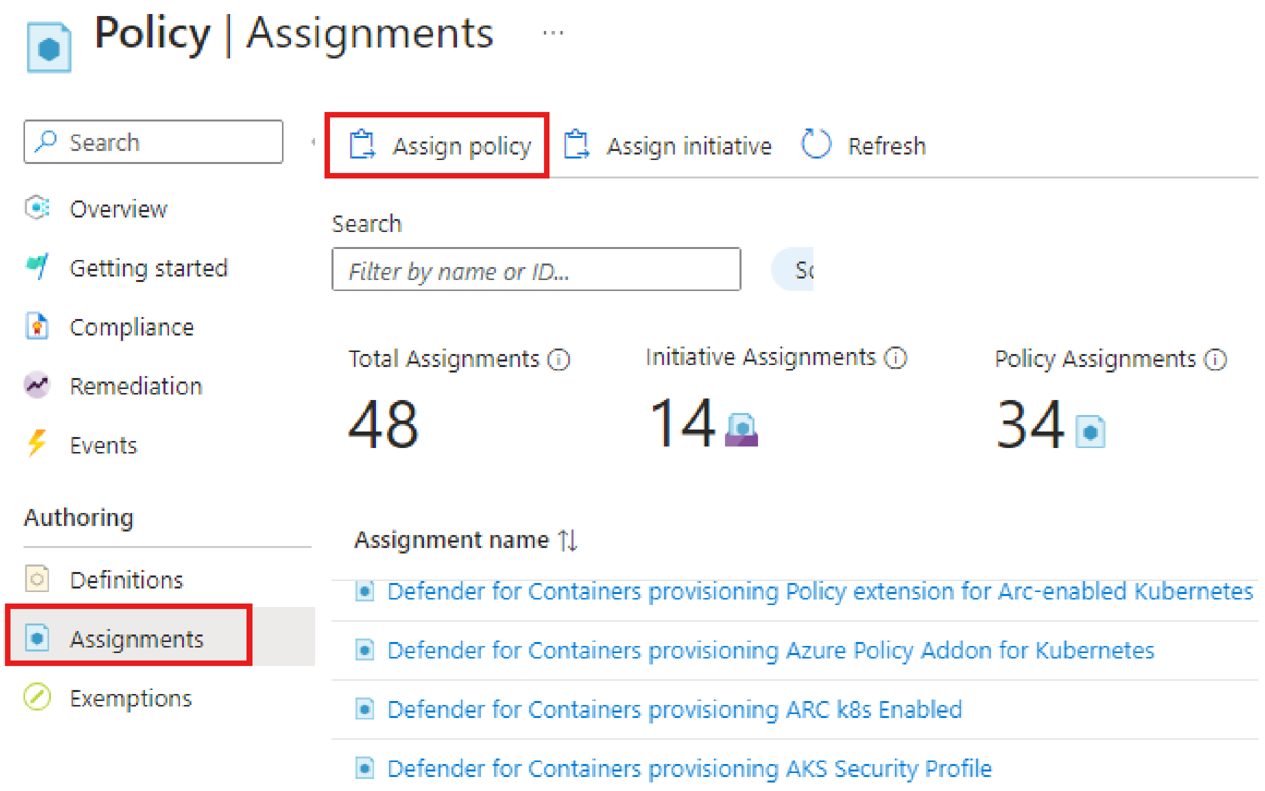
Demonstration – Azure Policy

- Assign a policy
- Create and assign an initiative definition
- Check for compliance
- Check for remediation tasks
- Remove your policy and initiative



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Secure your Azure resources with Azure role-based access control (Azure RBAC)

Learning Objective – Azure RBAC

- Compare Azure RBAC Roles to Entra ID Roles
- Create a Role Definition
- Create a Role Assignment
- Apply RBAC Authentication
- Demonstration – Azure RBAC
- Learning Recap

Exam objectives

Manage access to Azure resources

- Manage built-in Azure roles
- Assign roles at different scopes
- Interpret access assignments

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Compare Azure RBAC Roles to Entra ID Roles

Team Admin
Exchange Admin

RBAC roles provide fine-grained access management

Azure RBAC roles	Entra ID roles
Manage access to Azure resources	Manage access to Entra ID objects
Scope can be specified at multiple levels	Scope is at the tenant level
Role information can be accessed in the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager templates, REST API	Role information can be accessed in Azure portal, Microsoft 365 admin portal, Microsoft Graph PowerShell

Owner
Contrib.
Reader
User Access Admin
+ custom

Global Admin
User Admin
....
+ custom

✓ There are many built-in roles, or you can create your own custom role

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Create a Role Definition

Collection of permissions that lists the operations that can be performed

Owner
Contributor
Reader
…
Backup Operator
Security Reader
User Access Administrator
Virtual Machine Contributor

Built-in

Reader Support Tickets
Virtual Machine Operator

Custom

Contributor

```
"Actions": [
 "*"
],
"NotActions" : [
 "Authorization/*/Delete",
 "Authorization/*/Write",
 "Authorization/elevateAccess/Action"
],
"DataActions" : [],
 "NotDataActions": [],
 "AssignableScopes" : [
 "/"
]
```

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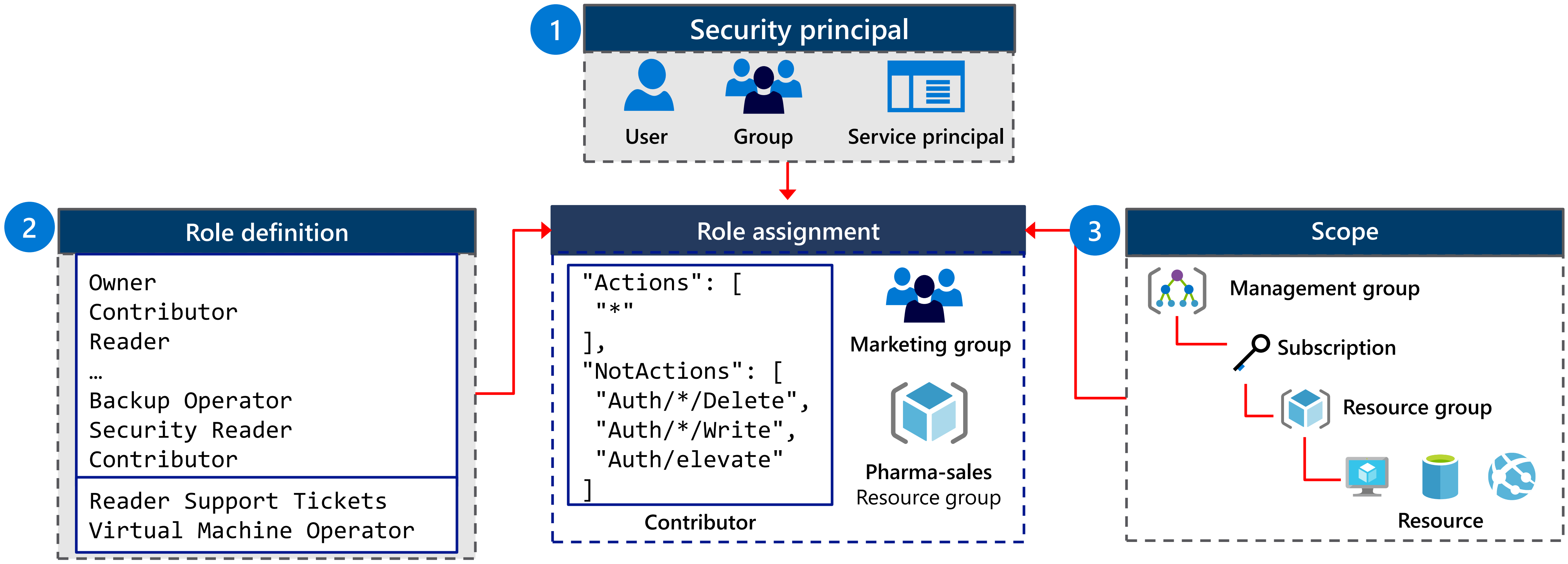
Create a Role Assignment

Process of binding a role definition to a user, group, or service principal at a scope for the purpose of granting access



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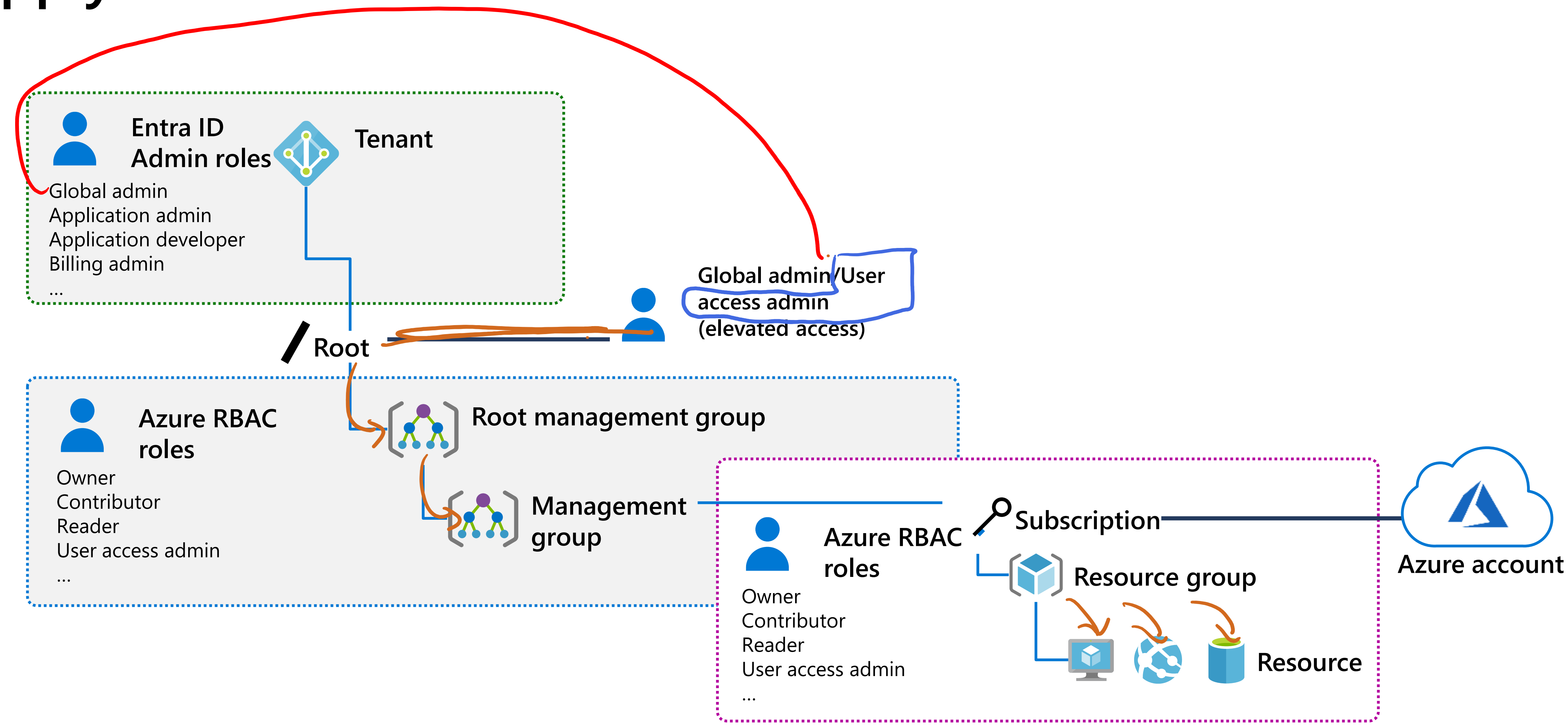
Apply RBAC Authentication



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Demonstration – Azure RBAC

- Locate the Access Control blade
- Review role permissions
- Add a role assignment

Add role assignment ···

Role • Members • Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles.
Assignment type

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Name ↑↓	Description ↑↓	Type ↑↓
Reader	View all resources, but does not allow you to make any ch...	BuiltInRole
Access Review Operator Service Role	Lets you grant Access Review System app permissions to ...	BuiltInRole
AcrDelete	acr delete	BuiltInRole
AcrImageSigner	acr image signer	BuiltInRole

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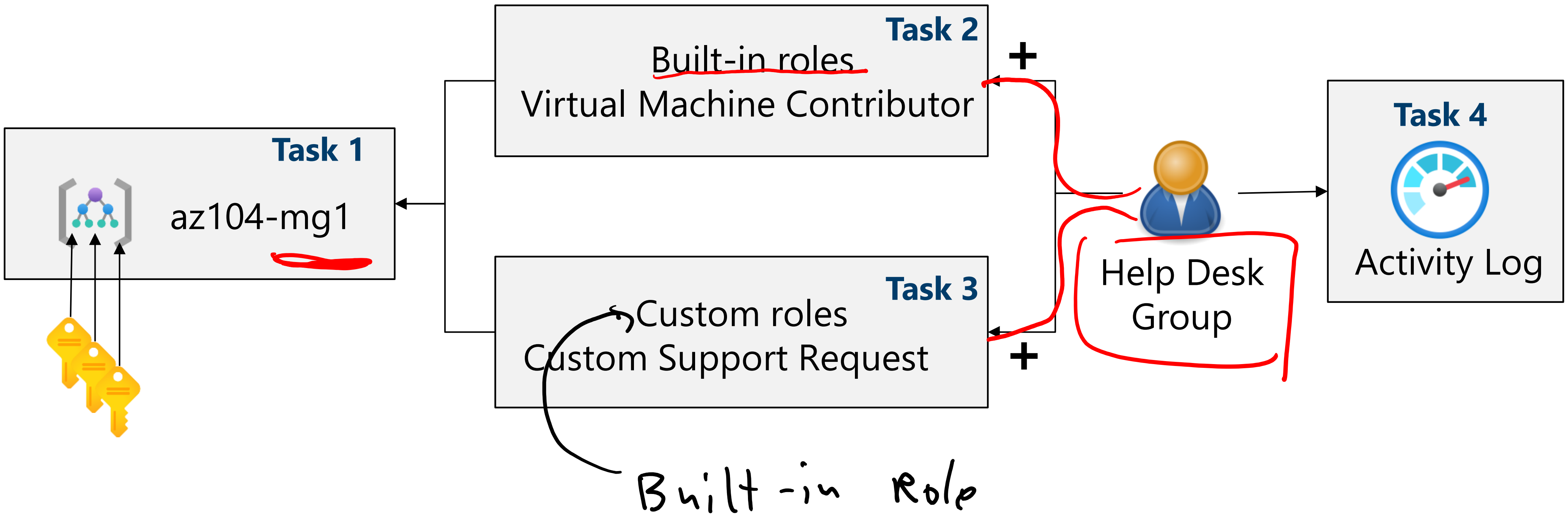
Lab 02a - Manage Subscriptions and RBAC

Lab 02b - Manage Governance via Azure Policy

Lab 02a – Architecture diagram



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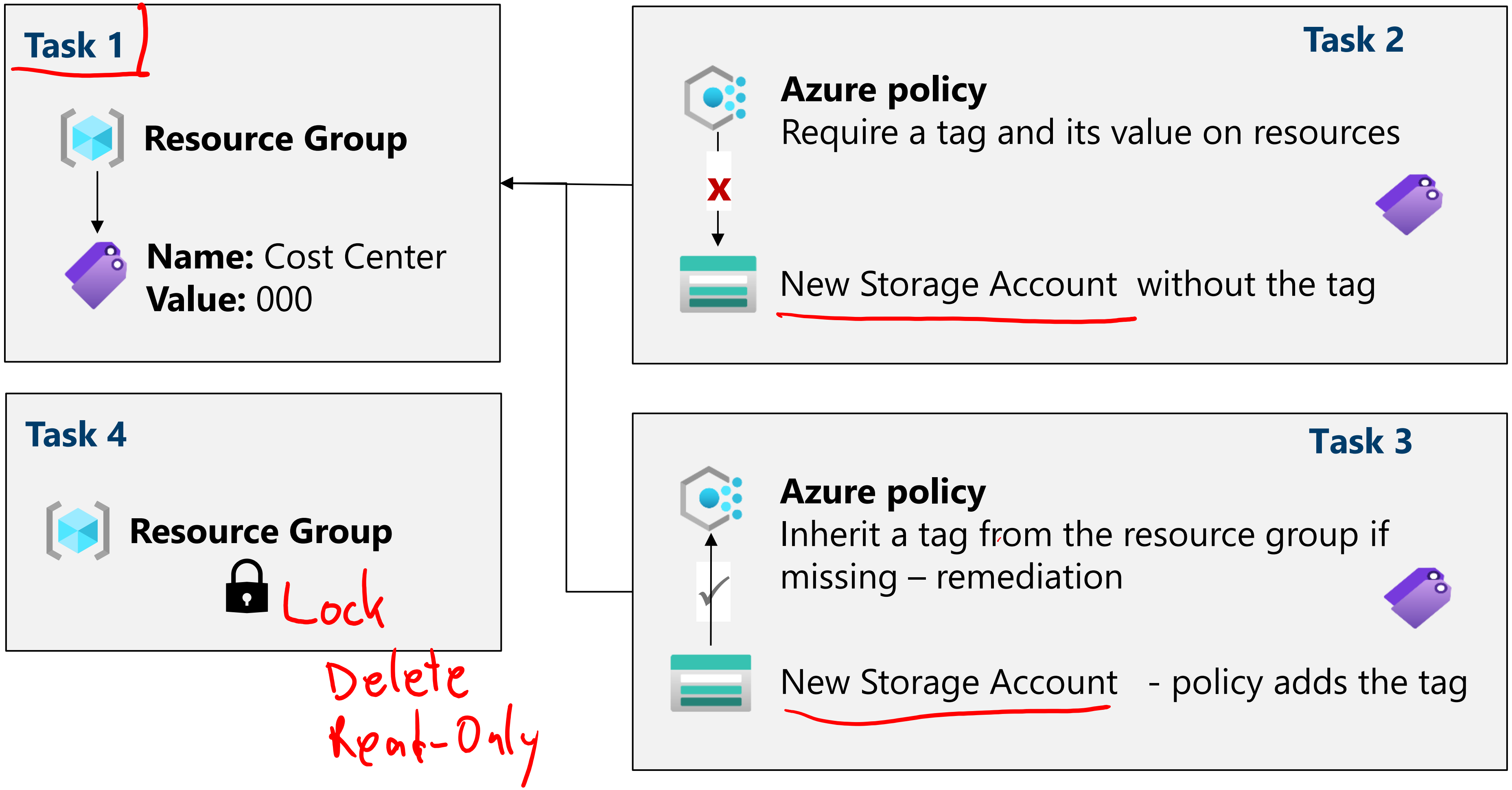
Lab 02b – Architecture diagram



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End of presentation